

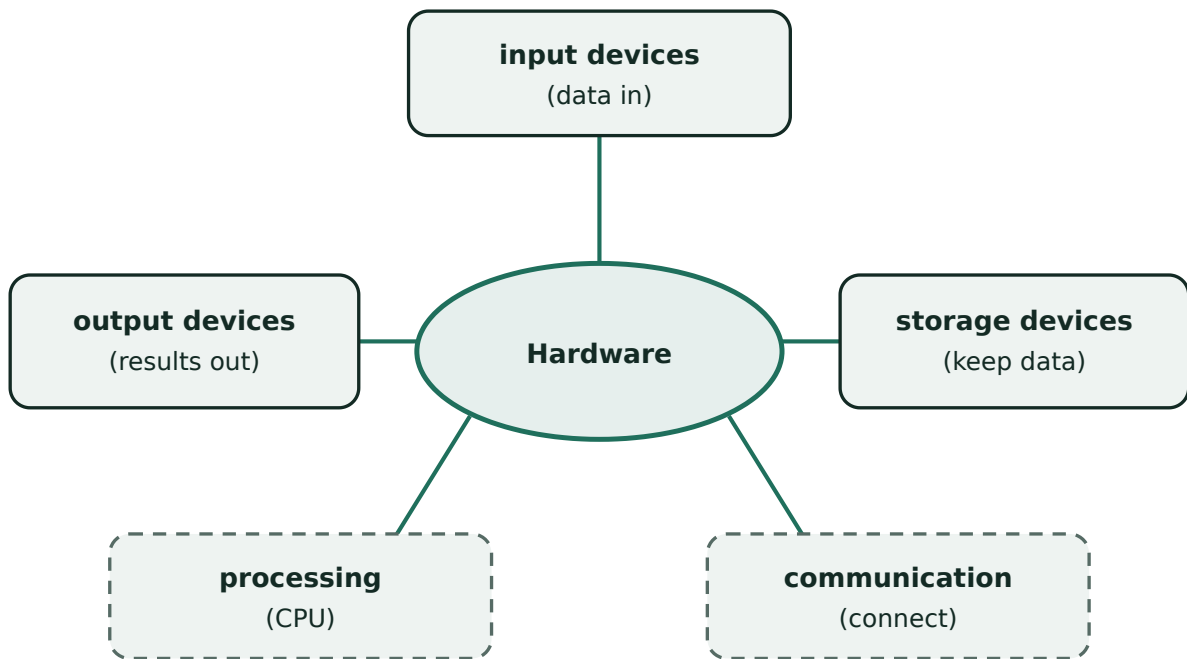
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Topic 2.1 - Input, output, and storage devices

A real classroom situation

The computer lab at a government high school in Larkana receives a donation: three boxes of equipment with no labels. The lab in-charge, Mr. Rashid, asks his B.Ed trainee, Miss Kiran, to sort the contents before the period ends. Out come a keyboard, a mouse, a pair of headphones, a flat-screen monitor, a bundle of USB flash drives, a small printer, a webcam still in its plastic, and a box of blank DVDs. Kiran realises she does not need to know each item's brand or price. She needs one piece of knowledge: what job each device does in the system. Sorted by job, the whole box organises itself.



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Explanation

Input devices

An input device puts data into the computer. The keyboard and mouse remain the primary input devices of a desktop (Bourgeois et al., 2019), but the family is much larger. A scanner turns a printed page into an image. A microphone turns voice into sound data. A webcam turns light into video. A biometric attendance machine reads a fingerprint. In a phone, the touchscreen is both an input device (your taps) and an output device (the display), which is worth noticing: classification follows the job being done at the moment, not the object.

Output devices

An output device brings results out of the computer in a form a human can use. The most obvious output device is the display or monitor; speakers and printers complete the usual set (Bourgeois et al., 2019). A projector in a classroom is an output device. So is the small screen of the biometric machine that shows a tick when your fingerprint is accepted.

Storage devices

A storage device keeps data for later, whether the power is on or off. The hard disk inside a desktop, the flash storage inside a phone, the USB flash drive in a pocket, the DVD in a drawer: all storage. Bourgeois and colleagues (2019) describe the difference plainly: a hard disk is non-volatile storage, meaning data survives when the computer is switched off, while the working memory the computer uses while running does not (we meet that memory in Topic 2.2). A solid state drive (SSD) does the same job as a hard disk with no moving parts, which makes it faster and more tolerant of being carried around.

Common devices in a school lab

Read across each row: device, family, function, and one school use.

Device	Family	What it does	One school use
Keyboard	Input	Types text and commands	Typing exam questions
Mouse	Input	Points and clicks	Navigating the desktop
Scanner	Input	Converts paper to image	Saving old records
Webcam	Input	Captures video	Recording lesson practice
Monitor	Output	Displays results	Showing a lesson video
Printer	Output	Prints on paper	Printing parent notices
Speakers	Output	Plays sound	Language listening lab
USB flash drive	Storage	Keeps files portable	Carrying marks sheets

Family names are written in every row, so the table remains clear without colour.

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A caution about the boundary cases, because they are exactly the cases a teacher meets. A touchscreen is input and output at once. A printer that scans is an output device in one minute and an input device the next. A USB flash drive is storage, but the USB port it plugs into belongs to the computer's connection system. When you classify, name the job first, then the family. The device detective habit from Unit 1 still applies.

One more distinction keeps the vocabulary clean: **peripheral**. A peripheral is any device attached to the computer rather than built into it. The keyboard,

printer and flash drive are peripherals; the storage chip inside the phone is not. The word describes attachment, not function.

Activity: Sort the donation

Work in pairs or threes. About 20 minutes.

1. Your teacher reads out a donation list of twelve items: keyboard, mouse, monitor, headphones, laser printer, scanner, webcam, USB flash drive, external hard drive, box of DVDs, interactive whiteboard, and a barcode reader from a library.
2. For each item, agree on: the family (input, output, or storage), the job it does in one clause, and one place it could be used in a school.
3. Two items on the list can belong to more than one family depending on use. Identify them and write one sentence for each explaining the dual role.
4. Compare your table with another group. Argue any disagreement by pointing to the job, not the object.

Check your understanding

1. Define an input device and an output device in one sentence each.
(*Remember*)
2. A colleague says a printer is "just an output device". Describe one situation in which that same physical machine is an input device. (*Understand*)
3. Your school keeps pupil records on paper and wants to digitise them. Name the three devices the job needs and state each one's family and role.
(*Apply*)
4. Explain why a USB flash drive is a storage device while the touchscreen of a phone is neither storage nor only input. (*Understand*)

Summary

Hardware sorts by job. Input devices put data in (keyboard, mouse, scanner, microphone, webcam, touchscreen); output devices bring results out (monitor, projector, speakers, printer); storage devices keep data whether or not the

power is on (hard disk, SSD, flash drive, DVD). Classify by the job being done at the moment, not by the object, and the boundary cases - touchscreens, scanner-printers - stop being confusing and start being interesting.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name at least four input devices, four output devices and four storage devices.
- ☐ I can classify a described device and justify the classification by its job.
- ☐ I can explain why some devices belong to two families depending on use.

Try this at your practicum school

List every hardware device in your practicum school's computer lab or office (aim for at least ten). Beside each, write its family (input, output, storage, or more than one) and one task it is used for. Note any device whose family surprised you.

Summative task

Your school receives a grant to equip a new "digital corner" in the library for pupil use. Prepare a one-page equipment plan: choose six devices, and for each give the device name, its family, its job in the digital corner, and one sentence justifying why it was chosen over an alternative (for example, headphones over speakers).

Marking guide (10 marks):

- Six devices chosen, each with a correct family label: 3 marks.
- Each device's job stated accurately in its own words: 2 marks.
- Justifications compare against an alternative using the device's actual capability (Analyse): 4 marks.
- Plan is clear and fits one page: 1 mark.

Further reading

- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 2. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Shelly, G. B., & Vermaat, M. E. *Discovering Computers*. Cengage Learning. (Course guide reading; print monograph, cited at bibliographic level.)
- Khan Academy. *Computers and the Internet*.
<https://www.khanacademy.org/computing/computers-and-the-internet>

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